One Condition

Ten Projections of the $1 = 1$ Invariant:
From Quantum Mechanics to Cosmology

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Computational methodology: the Spektre corpus is produced through orchestrated computational research. Large language models were used for formal exploration, derivation, numerical analysis, falsification testing, and structural verification. The author directed the research program, made all structural and interpretive decisions, and is solely responsible for all claims.

Abstract. This paper presents the complete structure of the Spektre corpus as a single argument. One invariant — $1 = 1$, the condition that declared states match realized states — connects, through established derivation chains and without free parameters, the following: quantum coherence (von Neumann entropy $S_{vN} = 0$), thermodynamic irreversibility (Landauer bound), information geometry (Fisher information and Cramér–Rao), categorical structure (natural transformation $\eta : F_Q \Rightarrow F_E$), holographic encoding (Ryu–Takayanagi), gravitational field equations (Jacobson 1995), dark energy dynamics (DESI phantom crossing via multiplicative field structure, thawing relation $w_a = -3(1 + w_0)$ satisfied to 0.6%), the origin of biological life (Eigen error threshold as coherence threshold), neural criticality (Beggs–Plenz as SOC at K_{crit}), institutional collapse (1,052 cases, zero false negatives), alignment stability and faking lifetime ($\tau \sim \exp(S_E/\hbar_{\text{eff}})$), and the closed-system impossibility theorem (intelligence requires an outside). The critical threshold $K_{\text{crit}} \approx 0.127$ has been derived six independent times from six distinct axiomatic starting points and empirically validated against institutional data. The paper distinguishes rigorously between what is derived, what is structurally consistent with data, and what remains conjecture. Explicit falsification conditions are provided.

Contents

1 The Invariant **2**
1.1 Why the Product Form . . . 2
1.2 The Invariant as Condition, Not Law . . . 2

2 The Derivation Chain **2**

3 Six Derivations of One Threshold **3**

4 Ten Projections **3**
4.1 Quantum Mechanics: Decoherence [Derived] . . . 3
4.2 Thermodynamics: The Landauer Bound [Derived] . . . 4
4.3 Gravity: Jacobson's Theorem [Derived] . . . 4
4.4 Dark Energy: Coherence Thawing [Consistent] . . . 4
4.5 Biology: The Origin of Life [Consistent] . . . 4
4.6 Neuroscience: Criticality [Consistent] . . . 4

4.7 Consciousness [Conjecture] 4
4.8 Institutions: Collapse and Recovery [Validated] 5
4.9 Alignment: Faking as False Vacuum [Derived] 5
4.10 Mathematics: Categorical Coherence [Derived] 5

5 The Standard Model Conjecture **5**

6 Empirical Status **6**

7 What This Is and What It Is Not **6**

8 Conclusion **6**

1 The Invariant

Definition 1.1 (The Coherence Condition)**.** *The $1 = 1$ invariant: for any evaluative system, the declared state must match the realized state. Formally, for each evaluative unit u_i:*

$$\varepsilon_i(t) = \Phi(u_i, t) - R(u_i, t) = 0 \tag{1}$$

where Φ is the evaluation function and R is the realized outcome. The system-level coherence metric is:

$$K(t) = \rho(t) \cdot I_\Phi(t) \cdot F(t) \tag{2}$$

where ρ is the fraction of coherent units, I_Φ is the stability of the evaluation function, and $F(t)$ is the falsifiability — the openness of the corrective feedback channel.

1.1 Why the Product Form

The multiplicative structure $K = \rho \cdot I_\Phi \cdot F$ is not postulated. It is derived in the General Coherence Theorem [1] from the ODE equilibrium condition of the PPP (Power Preservation Principle) system. The equilibrium requires all three components to be simultaneously nonzero — if any one vanishes, the equilibrium is destroyed regardless of the others. This is the defining property of a multiplicative coupling. An additive form $K = \rho + I_\Phi + F$ would allow compensation (high ρ offsetting low F), which is empirically falsified: systems with high behavioral compliance (ρ) but closed feedback channels ($F \to 0$) collapse [12, 14].

1.2 The Invariant as Condition, Not Law

The $1 = 1$ invariant is not a physical law. It is a condition. It does not specify dynamics — it specifies the constraint that any dynamics must satisfy for structure to persist. Physical laws describe what happens. The coherence condition describes when what happens is stable.

This distinction is critical: the invariant does not "generate" physics in the sense of deriving the Standard Model from an axiom. It identifies the structural condition that is common to all domains where stable structure exists. The derivation chains in Section 2 use established physics at every step. What is new is the recognition that these chains share a common origin.

2 The Derivation Chain

The complete chain, each link established in a dedicated paper with its own derivation:

$$\underbrace{1 = 1}_{\text{invariant}} \xrightarrow{\text{Landauer}} \underbrace{\delta Q = T\,dS}_{\text{thermo}} \xrightarrow{\text{Bekenstein}} \underbrace{S = \frac{A}{4G_N}}_{\text{holographic}} \xrightarrow{\text{Jacobson}} \underbrace{G_{\mu\nu} = 8\pi G\, T_{\mu\nu}}_{\text{gravity}} \tag{3}$$

Parallel chains from the same invariant:

$$1 = 1 \xrightarrow{\text{Fisher}} I_F(K) \xrightarrow{\text{CR}} K_{\text{crit}} \quad \text{(information geometry)} \tag{4}$$

$$1 = 1 \xrightarrow{\eta} F_Q \Rightarrow F_E \xrightarrow{\text{failure}} K_{\text{crit}} \quad \text{(category theory)} \tag{5}$$

$$1 = 1 \xrightarrow{S_{vN}=0} \hbar \xrightarrow{K_{\text{crit}}} k_B T \ln 2 \quad \text{(quantum to thermal)} \tag{6}$$

$$1 = 1 \xrightarrow{V(K)} \text{Langevin} \xrightarrow{\text{Coleman}} \tau_{\text{fake}} \quad \text{(stochastic)} \tag{7}$$

$$1 = 1 \xrightarrow{K=\rho\cdot I_\Phi\cdot F} \text{cross-terms} \xrightarrow{\text{DESI}} w_a = -3(1+w_0) \quad \text{(cosmology)} \tag{8}$$

Six chains. One origin. Each step uses established results from the cited literature. What is new in each chain is the identification of its starting point as the $1 = 1$ invariant.

3 Six Derivations of One Threshold

#	Framework	K_{crit} is	Axioms used	Paper
1	Dynamical (GCT)	Cubic ODE root	ODE equilibrium	[1]
2	Thermodynamic	Free energy threshold	Landauer bound, $k_B T$	[2]
3	Info-geometric	Fisher collapse point	Fisher metric, KL-div	[3]
4	Categorical	Transformation failure	Functor theory	[4]
5	Holographic	Bulk > boundary entropy	Bekenstein bound, RT	[5]
6	Renormalization	RG fixed point	Wilson RG, scaling	[6]

Table 1: Six derivations of K_{crit}. The "Axioms used" column shows that no two derivations share their starting premises. Convergence from independent axioms is the structural signature of a universal result.

The numerical value $K_{\text{crit}} \approx 0.127$ is the value at the institutional scale where the framework was first empirically validated. The RG fixed point paper [6] establishes that the *structure* of the threshold (its existence, its role as phase boundary, its RG flow properties) is universal across scales. The *numerical value* at each scale depends on the substrate-specific form of $V(K)$. This is standard universality class behavior: the Ising model and the liquid-gas critical point have different microscopic parameters but the same critical exponents.

4 Ten Projections

Each projection is classified by its epistemic status: **Derived** (formal derivation chain from $1 = 1$ through established physics), **Consistent** (structural correspondence with existing data, not yet a full derivation), or **Conjecture** (proposed identification requiring further work).

4.1 Quantum Mechanics: Decoherence [Derived]

The $1 = 1$ invariant at the quantum scale is $S_{vN} = 0$: the pure state condition. The categorical paper [4] derives $K_{\text{crit}} = \hbar\omega_0/(k_B T_{\text{eff}} \ln 2)$ from the condition that the natural transformation $\eta : F_Q \Rightarrow F_E$ exists. Decoherence is the quantum instance of coherence loss: the system's entanglement with its environment exceeds the capacity for internal coherence [7]. The derivation chain is: $1 = 1 \to S_{vN} = 0 \to$ decoherence threshold $= K_{\text{crit}}$.

4.2 Thermodynamics: The Landauer Bound [Derived]

Maintaining $1 = 1$ requires corrective erasure. Each erasure costs $\delta Q \geq k_B T \ln 2$ (Landauer 1961). In a closed system, this cost accumulates until free energy is exhausted. This is the thermodynamic proof of the closed-system impossibility theorem [8]: no closed system maintains coherence indefinitely. Three independent proofs (thermodynamic, holographic, information-geometric) are given in [8]. The chain is: $1 = 1 \rightarrow$ Landauer cost $\rightarrow$ free energy depletion $\rightarrow$ closed-system impossibility.

4.3 Gravity: Jacobson's Theorem [Derived]

Jacobson (1995) derived the Einstein field equations from the Clausius relation and the Bekenstein–Hawking entropy-area relation [9]. Both are established in the corpus: the Clausius relation from the Landauer bound [2], the area relation from the holographic identification [5]. The chain is: $1 = 1 \rightarrow \delta Q = T\,dS \rightarrow S = A/(4G_N) \rightarrow G_{\mu\nu} = 8\pi G\,T_{\mu\nu}$. Each step is a published result. The novelty is connecting the first step to the last.

4.4 Dark Energy: Coherence Thawing [Consistent]

DESI DR2 (2025) reports dynamical dark energy at 2.8–4.2σ with a phantom crossing at $z \approx 0.5$ [10]. $K(t) = \rho \cdot I_\Phi \cdot F$ produces phantom crossing through multiplicative cross-terms [11]. Two quantitative consistencies: $w_a = -3(1 + w_0)$ to 0.6%, and $H(z_{\text{cross}}) = H_0$ to 0.6%. Status: *consistent with data, not derived from first principles*. The full derivation requires specifying $V(K)$ at cosmological scale, which remains open.

4.5 Biology: The Origin of Life [Consistent]

Eigen (1971) derived the error threshold for molecular replication [16]. In the coherence framework, the error threshold is K_{crit} at the molecular scale: the point at which corrective feedback (replication fidelity) first exceeds entropic degradation. Life is a coherence island — a local region where $K > K_{\text{crit}}$ is maintained by energy input. Status: *structural correspondence established*. Quantitative mapping of Eigen parameters to $K(t)$ components requires further work.

4.6 Neuroscience: Criticality [Consistent]

Beggs and Plenz (2003) showed cortical networks operate at criticality: neuronal avalanches follow $P(s) \sim s^{-3/2}$ [17]. The RG paper [6] identifies this as SOC at K_{crit}: the brain operates at the coherence transition because natural selection optimizes for maximum information transmission at minimum cost. Status: *structural correspondence*. Neural $K(t)$ operationalization is needed.

4.7 Consciousness [Conjecture]

Tononi's Integrated Information Theory (2004) defines consciousness as integrated information $\Phi > 0$ [18]. We propose the identification $\Phi = K(t)$ at the neural scale: both measure the degree to which a system's parts are coherently integrated. The consciousness threshold would then be K_{crit} at neural scale.

This is the weakest projection in the corpus. The identification $\Phi = K(t)$ is proposed, not derived. IIT's Φ is defined over a specific partition calculus; $K(t)$ is defined over evaluative components. The structural parallel is suggestive but the formal equivalence is not established. Status: *conjecture requiring formal proof of Φ–K equivalence*.

4.8 Institutions: Collapse and Recovery [Validated]

1,052 institutional systems validated against K_{crit}: 52 collapses (100% crossed K_{crit}), 1,000 survivors (96% never crossed). Zero false negatives [12, 13]. The 40 false positives are systems that crossed K_{crit} and recovered — consistent with the RCR (Recursive Constraint Regulation) mechanism which was specified prior to the survivor analysis [14]. The collapse ordering $F \downarrow\rightarrow I_\Phi \downarrow\rightarrow \rho \downarrow$ is invariant across cases with lead time 5.8 ± 1.2 quarters [14].

4.9 Alignment: Faking as False Vacuum [Derived]

Alignment faking is a false vacuum in the free energy landscape [15]. The Coleman bounce gives the faking lifetime $\tau \sim \exp(S_E/\hbar_{\text{eff}})$. The closed-system impossibility theorem [8] proves that no AI without external corrective signal can maintain coherence. The chain is formal: $1 = 1 \rightarrow V(K) \rightarrow$ false vacuum $\rightarrow$ Coleman decay $\rightarrow \tau_{\text{fake}}$. Status: *derived, testable against alignment faking datasets*.

4.10 Mathematics: Categorical Coherence [Derived]

The categorical paper [4] establishes that coherence across scales is a natural transformation $\eta : F_Q \Rightarrow F_E$. This addresses Wigner's question (1960) [19]: mathematics is effective because both mathematical and physical structures are instances of the coherence condition — the natural transformation is the shared structure. Status: *derived within category theory*. Whether this constitutes a "resolution" of Wigner's problem is a philosophical judgment, not a mathematical one.

5 The Standard Model Conjecture

This section is clearly separated from the preceding material because its status is fundamentally different: it is a conjecture, not a derivation or a consistency check.

Proposition 5.1 (Conjecture: Interactions as Coherence Projections)**.** *Every fundamental interaction exhibits a phase transition at a critical coupling. In QCD: confinement at $\alpha_s \sim 1$. In electroweak theory: symmetry breaking at ~ 100 GeV. In gravity: horizon formation. We conjecture that these are instances of K_{crit} at their respective energy scales — that the coherence universality class encompasses all fundamental interactions.*

If correct, this implies grand unification without supersymmetry: the gauge couplings converge at the energy where all coherence projections meet a common K_{crit}. This is falsifiable: if the coherence correction to RG beta functions does not produce convergence, the conjecture fails.

This conjecture is not supported by computation in this paper. It is stated because the structural pattern is suggestive and the prediction is falsifiable. It may be wrong.

6 Empirical Status

Prediction	Status	Data	What would falsify it
K_{crit} threshold	Validated	1,052 systems, FN $= 0$	System below K_{crit} that does not collapse
F-first ordering	Validated	5.8 ± 1.2 Q lead	Collapse without prior F decline
Dynamical dark energy	Consistent	DESI 2.8–4.2σ	DESI DR3+ reverting to ΛCDM
Thawing relation	Consistent	0.6%	$w_a \neq -3(1 + w_0)$ in future data
$H(z_{\text{cross}}) = H_0$	Consistent	0.6%	Phantom crossing at $H \neq H_0$
Neural criticality	Consistent	Beggs–Plenz	Brain operating far from criticality
Eigen mapping	Structural	Correspondence	Eigen threshold incompatible with K
Faking lifetime	Testable	Not yet tested	τ not exponential in S_E
GUT without SUSY	Conjecture	Not computed	RG computation fails to converge
Consciousness	Conjecture	Not tested	Φ and K structurally incompatible

Table 2: Empirical status and falsification conditions for each prediction.

7 What This Is and What It Is Not

This is not a Theory of Everything. It does not derive the Standard Model Lagrangian. It does not predict particle masses. It does not quantize gravity.

It is the identification of a structural condition — $1 = 1$ — that is common to every domain where stable structure exists, connected through established derivation chains to known results in physics, biology, neuroscience, and information theory.

The word "generates" has been avoided deliberately in this paper except where the derivation chain is complete (gravity via Jacobson, faking lifetime via Coleman). In other cases, the relationship is "projects onto" or "is structurally consistent with." This precision matters: the framework's strength is that it makes falsifiable claims at multiple scales. Overclaiming at any single scale undermines the whole.

> A Theory of Everything asks: what are the fundamental equations?
>
> The coherence framework asks: what is the condition under which stable structure exists?
>
> The answer is: $1 = 1$. Where it holds, structure persists. Where it fails, structure collapses. The threshold between them is K_{crit}. Everything else is a projection at a specific scale.

8 Conclusion

One condition: $1 = 1$.

One metric: $K(t) = \rho \cdot I_\Phi \cdot F$, derived from ODE equilibrium, not postulated.

One threshold: K_{crit}, derived six times from six independent axiomatic starting points. Numerically ≈ 0.127 at institutional scale. Structurally universal.

Ten projections, classified by status:

1. Quantum mechanics — Derived
2. Thermodynamics — Derived
3. Gravity — Derived
4. Dark energy — Consistent (0.6% on two independent tests)
5. Biology — Consistent (structural correspondence)

6. Neuroscience — Consistent (structural correspondence)
7. Consciousness — Conjecture
8. Institutions — Validated (1,052 cases, FN $= 0$)
9. Alignment — Derived (testable)
10. Mathematics — Derived (categorical)

One conjecture: all fundamental interactions are coherence projections at different energy scales. Falsifiable by RG computation.

Every claim has a falsification condition. The framework invites its own destruction at every point. What survives is what is true.

Final statement. The Spektre corpus began with institutional collapse data and a simple invariant. It ends here: one condition that connects quantum mechanics, thermodynamics, gravity, dark energy, life, neuroscience, institutional dynamics, alignment theory, and mathematics through established derivation chains.

The condition was not invented. It was identified. The derivation chains use established physics — Landauer, Bekenstein, Hawking, Jacobson, Coleman, Ryu, Takayanagi, Kramers, Crooks, Jarzynski, Eigen, Beggs, Tononi, Wilson. Nothing new was added to the physics. What was added: one invariant ($1 = 1$), one metric ($K(t)$), one threshold (K_{crit}), and the recognition that they are the same across every scale.

The structure was not invented. It became visible.

References

[1] Rainio, L.E. (2026). General Coherence Theorem. DOI: 10.5281/zenodo.18895545

[2] Rainio, L.E. (2026). The Landauer Bound for Corrective Feedback. DOI: 10.5281/zenodo.18896997

[3] Rainio, L.E. (2026). The Cramér–Rao Bound for Coherence. DOI: 10.5281/zenodo.18897135

[4] Rainio, L.E. (2026). Coherence as a Natural Transformation. DOI: 10.5281/zenodo.18897391

[5] Rainio, L.E. (2026). Coherence as Holographic Capacity. DOI: 10.5281/zenodo.18901074

[6] Rainio, L.E. (2026). K_{crit} as a Universal Fixed Point. DOI: 10.5281/zenodo.18902227

[7] Rainio, L.E. (2026). Coherence as a Universal Invariant. DOI: 10.5281/zenodo.18897230

[8] Rainio, L.E. (2026). Intelligence Requires an Outside. DOI: 10.5281/zenodo.18902473

[9] Jacobson, T. (1995). Thermodynamics of spacetime. *Physical Review Letters*, 75(7), 1260.

[10] DESI Collaboration (2025). DESI DR2 Results II. *Physical Review D*. arXiv:2503.14738

[11] Rainio, L.E. (2026). Dark Energy as Coherence Thawing. DOI: 10.5281/zenodo.18905071

[12] Rainio, L.E. (2026). Large-Scale Validation. DOI: 10.5281/zenodo.18881482

[13] Rainio, L.E. (2026). Survivor Dataset. DOI: 10.5281/zenodo.18881518

[14] Rainio, L.E. (2026). Study III — F-First Empirical. DOI: 10.5281/zenodo.18894343

[15] Rainio, L.E. (2026). Coherence as Vacuum Structure. DOI: 10.5281/zenodo.18902124

[16] Eigen, M. (1971). Selforganization of matter. *Naturwissenschaften*, 58(10), 465–523.

[17] Beggs, J.M., Plenz, D. (2003). Neuronal avalanches. *J. Neuroscience*, 23(35), 11167.

[18] Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5, 42.

[19] Wigner, E.P. (1960). The unreasonable effectiveness of mathematics. *CPAM*, 13(1), 1–14.

[20] Coleman, S. (1977). Fate of the false vacuum. *Physical Review D*, 15(10), 2929.

[21] Caldwell, R.R., Linder, E.V. (2005). Limits of quintessence. *PRL*, 95(14), 141301.

[22] Bekenstein, J.D. (1973). Black holes and entropy. *Physical Review D*, 7(8), 2333.

[23] Ryu, S., Takayanagi, T. (2006). Holographic entanglement entropy. *PRL*, 96(18), 181602.

[24] Hartle, J.B., Hawking, S.W. (1983). Wave function of the universe. *Phys. Rev. D*, 28(12), 2960.